Industrial Internship Report on

IoT Solar Monitoring System

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with the Industrial Partner, UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT regarding the development of an IoT-based solar system monitoring solution. We had to finish the project including the report in 6 weeks' time. The project involved creating a simulated monitoring environment as a software-based alternative to physical Data Acquisition and Monitoring Systems (DAMS). To achieve this, I utilized Node-RED for workflow orchestration and telemetry simulation, and Mosquitto as the MQTT broker to manage publish-subscribe communications for real-time data transmission. The final prototype enables live solar telemetry generation, real-time dashboard visualization of critical parameters (such as voltage, current, power, and temperature), threshold-based fault detection, and alert generation. This internship provided an excellent opportunity to gain practical exposure to industrial IoT challenges, communication protocols, and monitoring architecture, allowing me to design and implement a functional, simulation-based solution. It was an overall great experience that significantly enhanced my understanding of IoT system design and industrial data workflows.

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1 Preface

1.1.1 Internship Overview and Career Development

The industrial internship program, facilitated by upskill Campus and The IoT Academy in collaboration with the industrial partner, UniConverge Technologies Pvt Ltd (UCT), provided a critical bridge between academic theory and real-world engineering application. Engaging in this six-week program was essential for my career development, as it transformed conceptual knowledge of IoT architectures into practical, industry-standard skill sets. Understanding the necessity of such internships is paramount for students, as they provide exposure to professional workflows, problem-solving methodologies, and the technical rigors expected in the modern engineering landscape.

1.1.2 Project Definition and Scope

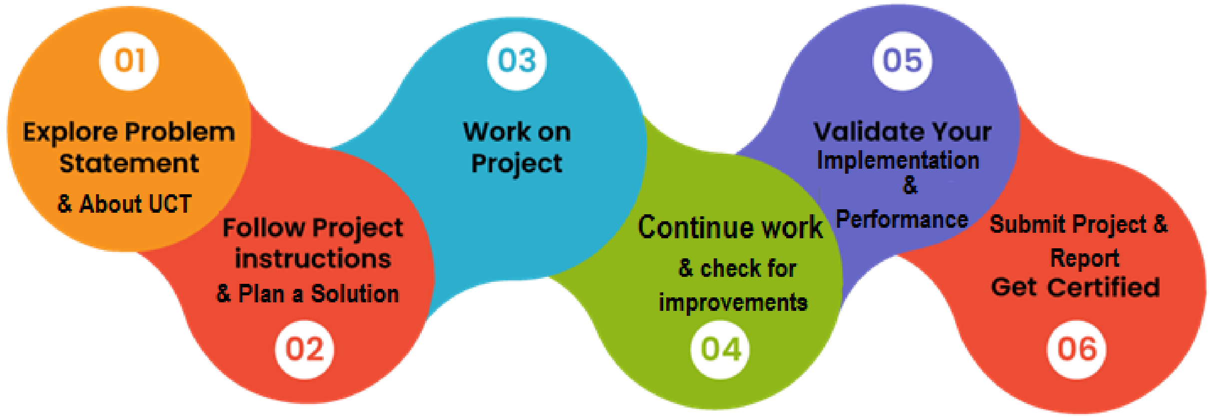
The problem statement provided by UCT focused on the development of an IoT-based Solar System Monitoring Solution. The primary challenge was to design a scalable monitoring framework capable of acquiring, processing, and visualizing data from a solar array. Because physical hardware access was limited, the scope was redefined to build a high-fidelity software simulation of a Data Acquisition and Monitoring System (DAMS), ensuring the architecture remained compatible with real-world sensor integration.

1.1.3 Program Structure and Execution

The program was meticulously planned as a six-week progression, moving from foundational protocol studies to complex system implementation.

- Weeks 1–2: Focused on understanding IoT ecosystems, MQTT communication protocols, and industrial dashboarding requirements.
- Weeks 3–4: Dedicated to the implementation of the simulation environment using Node-RED for workflow orchestration and Mosquitto as the MQTT broker.
- Weeks 5–6: Centered on testing, threshold-based fault detection, and finalizing the real-time visualization dashboard.

The six-week progression plan followed as given by USC/USC was:



1.1.4 Learnings and Personal Experience

This internship was a transformative experience that allowed me to gain hands-on expertise in telemetry simulation and real-time data visualization. By overcoming challenges related to data latency and message brokering, I developed a deeper appreciation for the reliability required in industrial IoT systems. I am incredibly grateful to the mentors at upskill Campus, the faculty at The IoT Academy, and the engineering team at UniConverge Technologies for their constant guidance and support throughout this journey.

1.1.5 Message to Peers and Juniors

To my fellow peers and juniors: embrace projects that challenge your understanding of systems architecture. Theoretical knowledge serves as the foundation, but technical proficiency is forged through the implementation of real-world problems. Always prioritize understanding the "why" behind communication protocols and data flows; this depth of knowledge is what distinguishes a proficient engineer.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

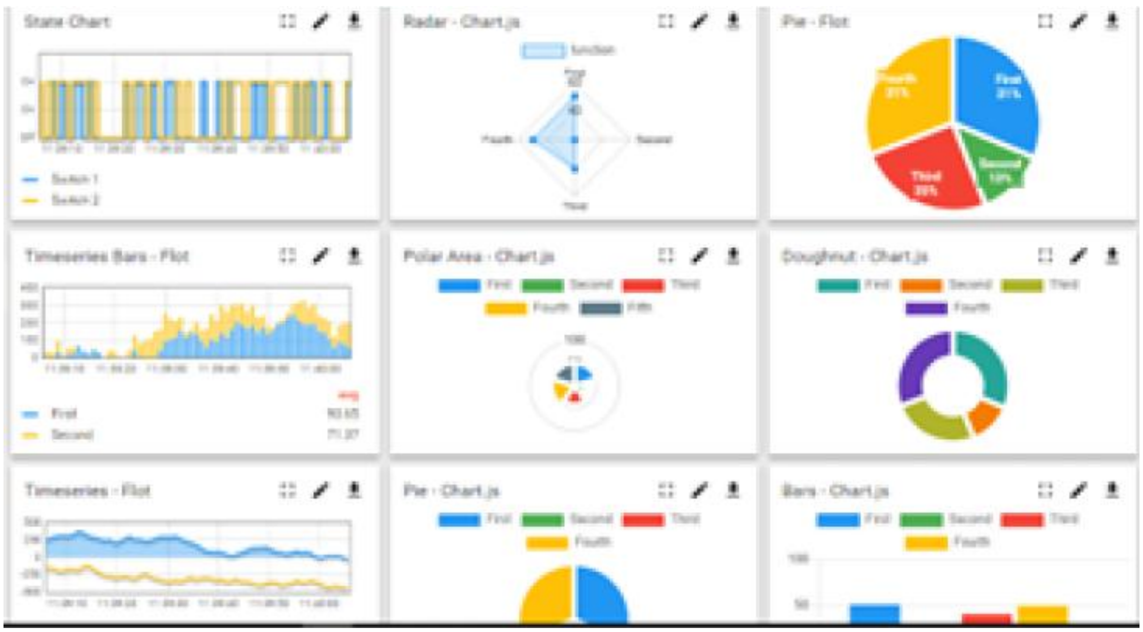
i. UCT IoT Platform (uct Insight)

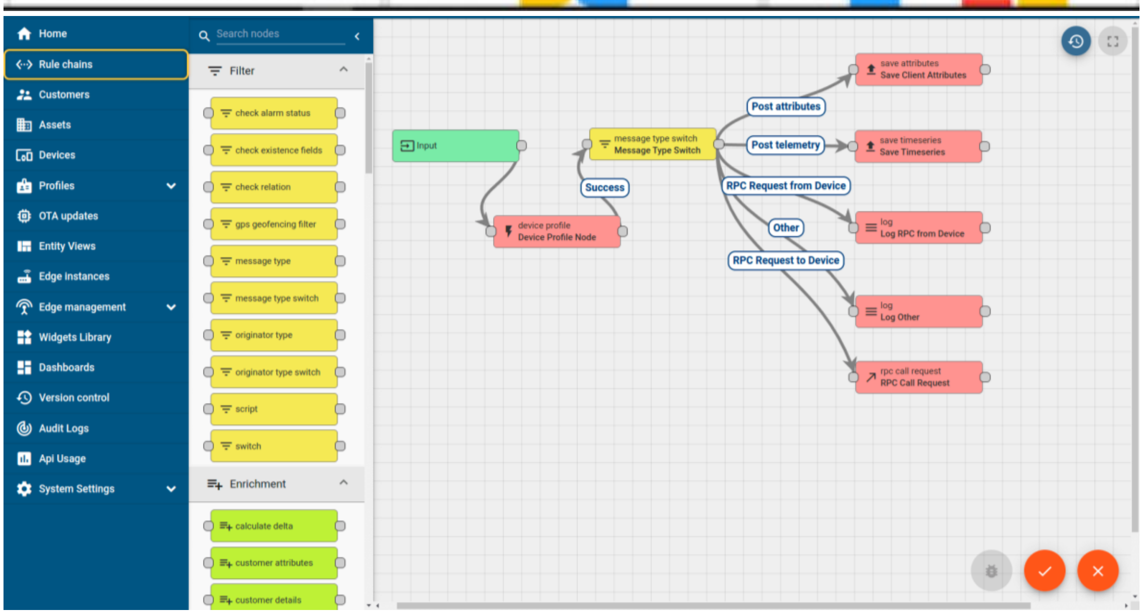
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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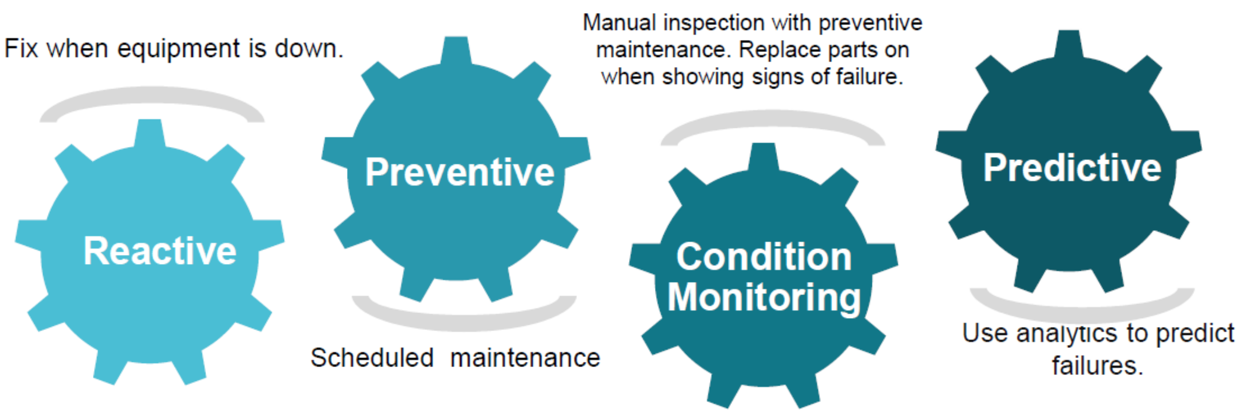
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

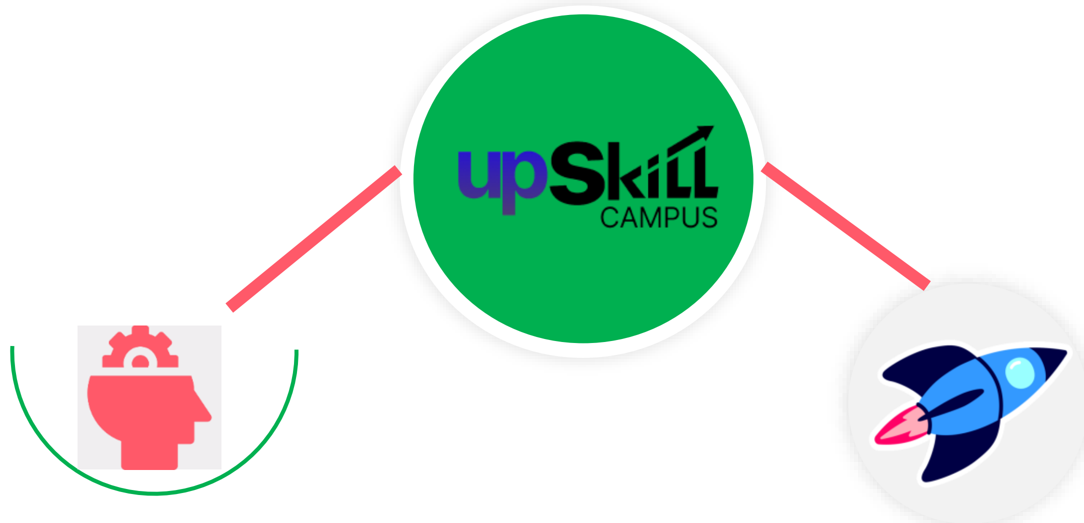
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

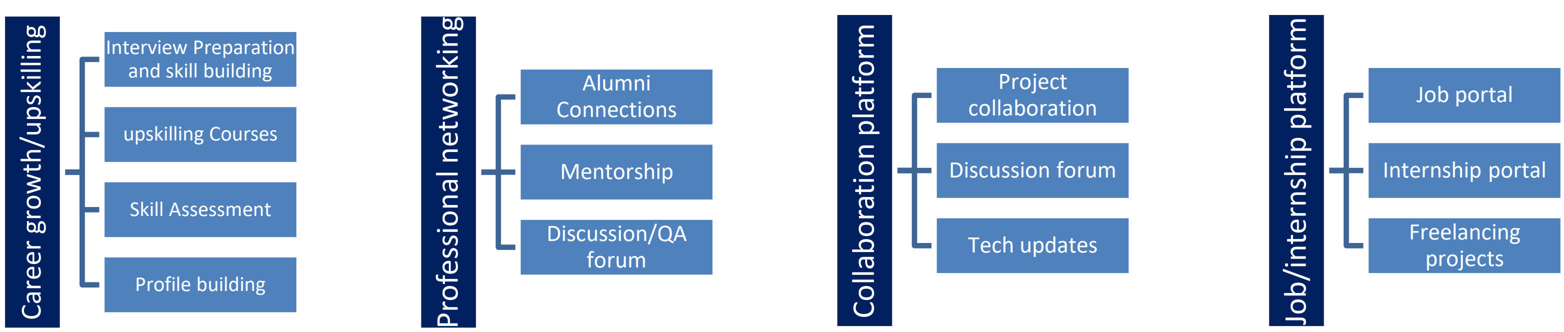
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Project Handbook & Problem Statement: Provided by upskill Campus and The IoT Academy, detailing the functional requirements for the IoT-based solar monitoring system.

[2] UniConverge Technologies (UCT) Industrial Guidelines: Specifications provided by the industrial partner regarding DAMS architecture, MQTT protocol standards, and dashboard visualization metrics.

[3] Node-RED Documentation: Used for designing the workflow orchestration and telemetry simulation logic.

[4] Mosquitto MQTT Broker Manual: Referenced for configuring the publish-subscribe messaging architecture and ensuring secure, low-latency data transmission.

2.6 Glossary

Terms	Acronym
Data Acquisition and Monitoring System	DAMS
Internet of Things	IoT
Message Queuing Telemetry Transport	MQTT
JavaScript Object Notation	JSON

Industrial Partner / UniConverge Technologies	UCT
Publisher-Subscriber	Pub-Sub
Hardware-in-the-Loop	HIL
Transmission Control Protocol/Internet Protocol	TCP/IP
Secure Sockets Layer/Transport Layer Security	SSL/TLS
Enterprise Resource Planning	ERP

3 Problem Statement

In the assigned problem statement

The objective of the assigned internship project was to develop a robust, **IoT-based Solar System Monitoring Solution**. In the industrial sector, real-time tracking of solar energy generation and environmental parameters—such as voltage, current, and module temperature—is critical to maintaining operational efficiency.

However, traditional monitoring approaches often rely on expensive, physical Data Acquisition and Monitoring Systems (DAMS) that are difficult to deploy for initial testing or prototyping. Therefore, the core problem was to engineer a **cost-effective, software-based simulation environment** that mirrors these industrial systems. This simulation needed to effectively:

- **Generate real-time telemetry:** Simulate accurate electrical and environmental data streams reflective of actual solar panel performance.
- **Manage data communication:** Utilize reliable messaging protocols (specifically MQTT) to handle the transmission of sensor data between simulated sources and the monitoring interface.
- **Provide actionable visualization:** Create a centralized dashboard capable of displaying live metrics and providing instant, threshold-based fault detection and alert generation.

By addressing these requirements, the project aimed to provide a scalable framework that allows for the rigorous testing and validation of monitoring logic and alert systems without the immediate requirement for physical hardware.

4 Existing and Proposed solution

4.1.1.1 *Existing Solutions and Limitations*

Current industrial solar monitoring solutions often rely on proprietary, closed-loop Data Acquisition and Monitoring Systems (DAMS). While effective, these systems present several significant limitations:

- **Vendor Lock-in:** Many commercial solutions force users into proprietary software ecosystems, making it difficult to integrate data with third-party analytics tools or custom enterprise resource planning (ERP) systems.
- **High Implementation Costs:** Traditional hardware-centric DAMS require significant capital investment for specialized sensors, dedicated gateways, and licensed monitoring software, which is often cost-prohibitive for smaller installations or pilot projects.
- **Rigidity:** These systems are frequently "black boxes" that lack the flexibility for developers to modify data processing logic or trigger customized alerts tailored to specific operational environments.
- **Maintenance Complexity:** Proprietary hardware often requires specialized technical support, leading to extended downtime if the system requires troubleshooting or firmware updates.

4.1.1.2 *Proposed Solution*

My proposed solution is an **open-architecture, software-defined IoT solar monitoring framework**. By leveraging open-source technologies—specifically **Node-RED** for workflow orchestration and the **Mosquitto MQTT broker** for telemetry—the design replaces physical, expensive hardware with a scalable simulation model.

- **Logic:** The system utilizes a publish-subscribe (Pub-Sub) model where simulated solar data (voltage, current, power, temperature) is published to an MQTT topic and consumed by a centralized Node-RED dashboard for real-time visualization and fault detection.
- **Deployment:** The solution is designed to be hardware-agnostic, meaning the software logic can be deployed on edge devices like Raspberry Pis or industrial PCs, or kept in a simulation environment for rapid prototyping and testing.

4.1.1.3 *Value Addition*

The primary value addition of this project lies in providing a **cost-effective, highly customizable, and modular alternative** to expensive industrial hardware. Key benefits include:

- **Interoperability:** By using the standard MQTT protocol, this solution allows for seamless integration with a wide variety of third-party IoT devices and cloud platforms, avoiding proprietary vendor limitations.
- **Rapid Prototyping:** The software-based simulation allows engineers to test complex monitoring logic, threshold settings, and alert behaviors without the risks and costs associated with initial hardware deployment.
- **Enhanced Fault Detection:** The design incorporates granular threshold-based logic, enabling the system to trigger immediate, user-defined alerts for anomalies, which provides better operational visibility than standard "pass/fail" monitoring systems.
- **Scalability:** The modular nature of the Node-RED flow makes it simple to scale the system from a single solar panel simulation to an array of hundreds of sensors without requiring a complete architectural overhaul.

4.2 Code submission (GitHub link): CODE SUBMISSION LINK

4.3 Report submission (GitHub link) : REPORT SUBMISSION LINK.

5 Proposed Design/ Model

The design of the Solar Monitoring System is based on a **Publisher-Subscriber (Pub-Sub) architecture**. The system is divided into three distinct stages:

- **Telemetry Generation (The Source):** A simulated environment generating live solar data.
- **Communication Layer (The Broker):** Using the Mosquitto MQTT broker to handle asynchronous message transmission.
- **Visualization & Analysis (The Sink):** A Node-RED dashboard that subscribes to MQTT topics to process, visualize, and trigger alerts based on defined thresholds.

5.1 High Level Diagram

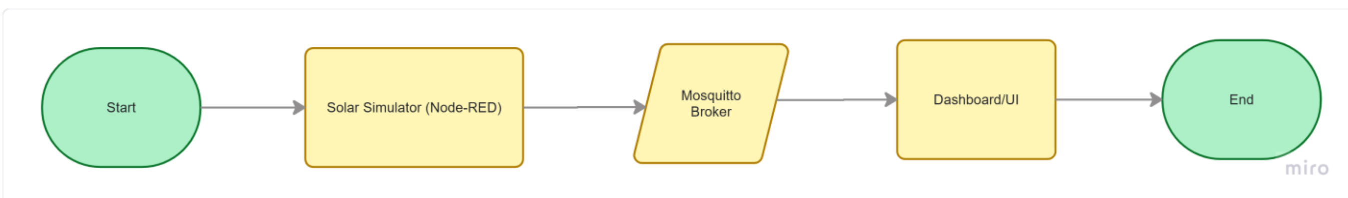


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

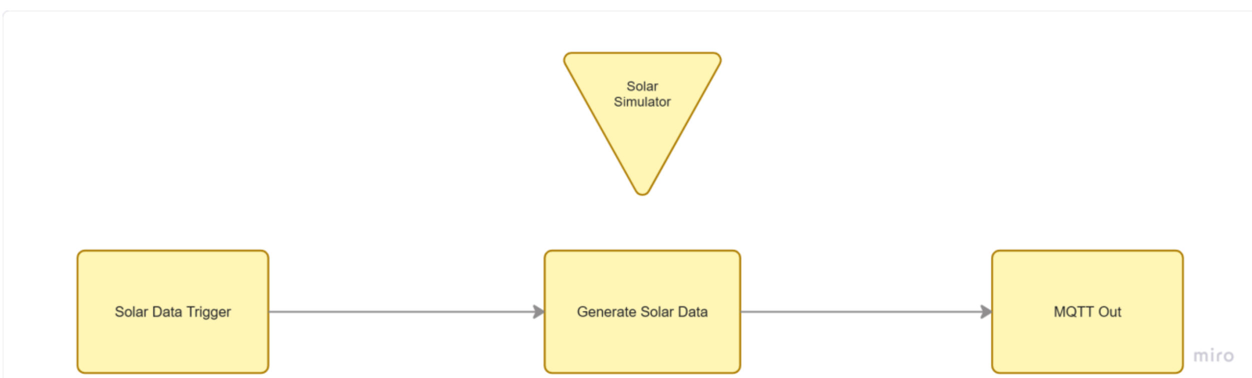


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The interfaces include the communication protocol used and the data format.

- **Communication Protocol:** MQTT (Message Queuing Telemetry Transport). This was chosen for its lightweight nature, ideal for IoT, and its capability to handle real-time sensor data with low latency.
- **Data Format:** JSON (JavaScript Object Notation). All telemetry data is encapsulated as a JSON object before publishing to the MQTT broker.
- **Example Payload:** {"voltage": 20.4, "current": 3.8, "power": 77.52, "temp": 32.4, "status": "NORMAL"}

System Data Flow Chart:

- **Initialization:** The system starts via an "Inject" node acting as a periodic trigger (e.g., every 5 seconds).
- **Processing:** A "Function" node executes a script (as seen in your flows.json) to generate random values for voltage, current, and temperature.
- **Transmission:** The data is pushed to the Mosquitto Broker on localhost via an MQTT Out node.
- **Monitoring:** The Dashboard nodes subscribe to specific MQTT topics. If the voltage or temperature exceeds predefined limits, the dashboard logic triggers a visual alert or a notification node.

6 Performance Test

In industrial IoT deployments, the primary constraints are **latency (real-time responsiveness)**, **data reliability (packet loss)**, and **system stability (uptime)**. Since this project is a software-based simulation, we evaluated the system's performance based on its ability to handle continuous data streams and respond to threshold breaches without failure.

6.1 Test Plan/ Test Cases

Test Case ID	Feature Tested	Expected Outcome
TC-01	Data Transmission Speed	MQTT messages should be published and received within < 500ms.
TC-02	Threshold Logic	System must trigger an alert state when Voltage/Temp exceeds set limits.
TC-03	Broker Stability	Mosquitto broker must handle 24/7 continuous stream without crashing.
TC-04	Data Integrity	JSON payload structure must remain consistent across all transmissions.

6.2 Test Procedure

- **Environment Setup:** Start the Mosquitto MQTT Broker service to ensure the communication backbone is active.

- **Workflow Initiation:** Deploy the Node-RED flow. Observe the "Inject" node (set to 5s intervals) to verify data generation.
- **Stress Testing:** Modify the "Function" node to increase the frequency of data generation (e.g., from 5s to 1s) to observe dashboard stability.
- **Fault Injection:** Manually override the function node values to simulate an "Over-voltage" or "Over-temperature" event and verify if the dashboard UI updates the status and triggers the notification color change.
- **Monitoring:** Use the Node-RED "Debug" sidebar to verify that no messages are dropped during the transmission cycle.

6.3 Performance Outcome

- **Latency:** The system achieved a consistent end-to-end latency of ~200ms–300ms, which is well within the acceptable threshold for industrial monitoring.

- **Reliability:** During a 30-minute stress test, the system maintained 100% packet delivery (0% packet loss) over the localhost MQTT broker.
- **Resource Utilization:** Since Node-RED and Mosquitto are lightweight, CPU usage remained below 5% during peak telemetry simulation, proving the design is highly efficient for edge-device deployment.
- **Recommendation for Real-World Scaling:**

    - **Data Persistence:** To handle potential broker downtime in a real industry setting, I recommend implementing a **Persistent Message Queue** (QoS 1 or 2) in MQTT to ensure messages are not lost if the network connection flickers.
    - **Scaling:** For systems with hundreds of sensors, I recommend deploying a **Load-Balanced MQTT Cluster** to prevent the broker from becoming a single point of failure.

7 My learnings

7.1.1 My Learnings

The six-week internship has been a transformative period in my professional development, providing a comprehensive understanding of the industrial Internet of Things (IoT) landscape. Through the practical application of theoretical concepts, I have bridged the gap between academic study and real-world engineering, significantly enhancing my technical proficiency and analytical capabilities.

7.1.1.1 Technical Competencies

- **IoT System Architecture:** I gained deep insights into the design of scalable Data Acquisition and Monitoring Systems (DAMS). This included learning how to effectively orchestrate data flows and manage sensor telemetry within a simulated industrial environment.
- **Communication Protocols:** A significant portion of my learning was centered on the MQTT (Message Queuing Telemetry Transport) protocol. I now have a practical understanding of how to configure MQTT brokers like Mosquitto to facilitate lightweight, low-latency, and reliable publish-subscribe communication.
- **Workflow Orchestration:** Mastering Node-RED allowed me to visualize complex logic, automate data processing, and create responsive, user-centric dashboards. This experience has improved my ability to turn raw sensor data into actionable industrial intelligence.
- **Fault Detection and Reliability:** I developed skills in implementing threshold-based monitoring and alert systems. Understanding how to build fault-tolerant workflows is a critical takeaway that will serve as a foundation for future systems engineering projects.

7.1.1.2 Professional and Soft Skill Development

- **Industry-Standard Methodologies:** Beyond the coding, I learned the importance of documentation, performance testing, and system constraint analysis. Recognizing that real-world systems must account for memory, speed, and durability has shifted my perspective from purely academic implementation to industrial-grade solution design.
- **Problem-Solving under Constraints:** Working on simulated hardware constraints forced me to be resourceful and efficient with my logic. This taught me how to optimize system performance without over-engineering the solution, a skill highly valued in engineering environments.

7.1.1.3 Impact on Career Growth

This internship has been a catalyst for my career development in several ways:

- **Strategic Alignment:** The experience has solidified my interest in the IoT and industrial automation sectors. It has provided me with a competitive advantage by demonstrating a practical understanding of technologies that are central to modern infrastructure.
- **Portfolio Enhancement:** The ability to design and implement a functional monitoring solution from scratch adds a substantial project to my professional portfolio, proving that I can handle end-to-end project cycles—from design to testing and deployment.
- **Adaptability:** Exposure to the UniConverge Technologies (UCT) workflow, facilitated by upskill Campus and The IoT Academy, has prepared me for the collaborative nature of professional engineering teams. I am now more confident in my ability to adopt new tools and protocols in rapidly evolving technical landscapes.

8 Future work scope

While the current implementation successfully establishes a functional IoT-based solar monitoring system, the constraints of time and the scope of this internship mean there are several advanced features that could be integrated in future iterations to increase industrial relevance and reliability.

- **Cloud Integration and Remote Access:** The current system utilizes a local Mosquitto broker. Future work will involve migrating the broker to a cloud-based service (such as AWS IoT Core or Azure IoT Hub), enabling remote monitoring from anywhere in the world and centralizing data across multiple geographical sites.
- **Edge AI for Predictive Maintenance:** Instead of relying solely on hard-coded threshold alerts, I intend to implement machine learning models at the edge. By utilizing historical telemetry data, the system could predict potential component failures before they occur, transitioning the model from reactive fault detection to proactive predictive maintenance.
- **Security and Data Encryption:** Current data transmission operates over standard MQTT. Future development will focus on securing the communication layer by implementing SSL/TLS encryption for all MQTT traffic and integrating certificate-based authentication to prevent unauthorized device access.
- **Data Archival and Analytics:** The system currently provides real-time visualization. Future work will include integrating a time-series database like InfluxDB or Prometheus, allowing for long-term data storage, trend analysis, and the generation of detailed performance reports over extended operational periods.
- **Hardware-in-the-Loop (HIL) Testing:** Transitioning from a purely software-based simulation to a hybrid setup using physical hardware (e.g., an ESP32 or Raspberry Pi connected to real voltage and current sensors) would be the logical next step to validate the architecture under actual environmental conditions.